

2.3 Eukaryotic cell cycle and division

- i Know that the cell cycle is a regulated process in which cells divide into two identical daughter cells, and that this process consists of three main stages: interphase, mitosis and cytokinesis.
- ii Understand what happens to genetic material during the cell cycle, including the stages of mitosis.
- iii Understand how mitosis contributes to growth, repair and asexual reproduction.

CORE PRACTICAL 3: Make a temporary squash preparation of a root tip to show stages of mitosis in the meristem under the light microscope.

- iv Understand how meiosis results in haploid gametes, including the stages of meiosis.
- v Understand that meiosis results in genetic variation through recombination of alleles, including independent assortment and crossing over.
- vi Understand what chromosome mutations are, as illustrated by translocations.
- vii Understand how non-disjunction can lead to polysomy, including Down's syndrome, and monosomy, including Turner's syndrome.